Northern Devon Healthcare NHS
NHS Trust

Document Control

Title			
Pre-operative Anaemia Guideline (Elective major joint replacement surgery)			
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Lead Director Medical Director			
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Consulted with the following stakeholders: (list all) • Hospital Transfusion Committee • Pre-operative Assessment Lead Nurse • Pre-op Assessment Consultants • Orthopaedics Lead Clinician • Medicines Management Team • Clinical Pharmacy Manager			
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1. Purpose

1.1. The purpose of this document is to define the process for management of patients who are found to be anaemic prior to major joint replacement.

1.2. The guideline applies to pre-op assessment, anaesthetic, surgical, haematology, pharmacy and primary care staff.

1.3. Implementation of this guideline will ensure that:

- Patients undergoing major joint replacement are not put at avoidable perioperative risk due to untreated pre-operative anaemia

2. Definitions

2.1.	Hb	Haemoglobin level (g/L)
2.2.	CRP	C-reactive protein
2.3.	Anaemia	Male Hb < 130 g/L
		Female Hb < 120 g/L
2.4.	Iron deficiency	Anaemia with ferritin < 30 ug/L
2.5.	Functional iron deficiency	Anaemia with ferritin < 100 and elevated CRP

3. Responsibilities

3.1. Implementation of this guideline will be the responsibility of the Lead Clinician and Lead Nurse for Pre-operative assessment in liaison with the Hospital Transfusion Committee and other key stakeholders detailed below.

Role of Pre-op Assessment Nurse

3.2. The Pre-op Assessment Nurse is responsible for:

- Ensuring that patients with pre-operative anaemia are identified at the earliest possible opportunity
- Ensuring that patients with pre-operative anaemia are offered treatment as detailed in this guideline
- Ensuring that patients' GP practices are notified of any newly identified anaemia and of any initial treatment commenced or required
- Ensuring that patients treated for pre-operative anaemia have their haemoglobin level re-assessed prior to surgery

Role of the Pre-op Assessment Anaesthetist

3.3. The Pre-op Assessment Anaesthetist is responsible for:

- Providing specialist advice and support in the management of pre-operative anaemia

Role of Haematology Consultant

3.4. The Haematology Consultant is responsible for:

- Provide specialist advice and support in the management of pre-operative anaemia

4. Pre-operative anaemia management

4.1. See flowchart in Appendix 6.1

5. Monitoring Compliance with and the Effectiveness of the Guideline

Standards/ Key Performance Indicators

5.1. Key performance indicators include:

- Haemoglobin levels at pre-op assessment prior to major joint replacement
- Final haemoglobin levels prior to major joint replacement
- Transfusion rates in major joint replacement surgery
- Haemoglobin levels at discharge following major joint replacement
- Length of hospital stay following major joint replacement

Process for Implementation and Monitoring Compliance and Effectiveness

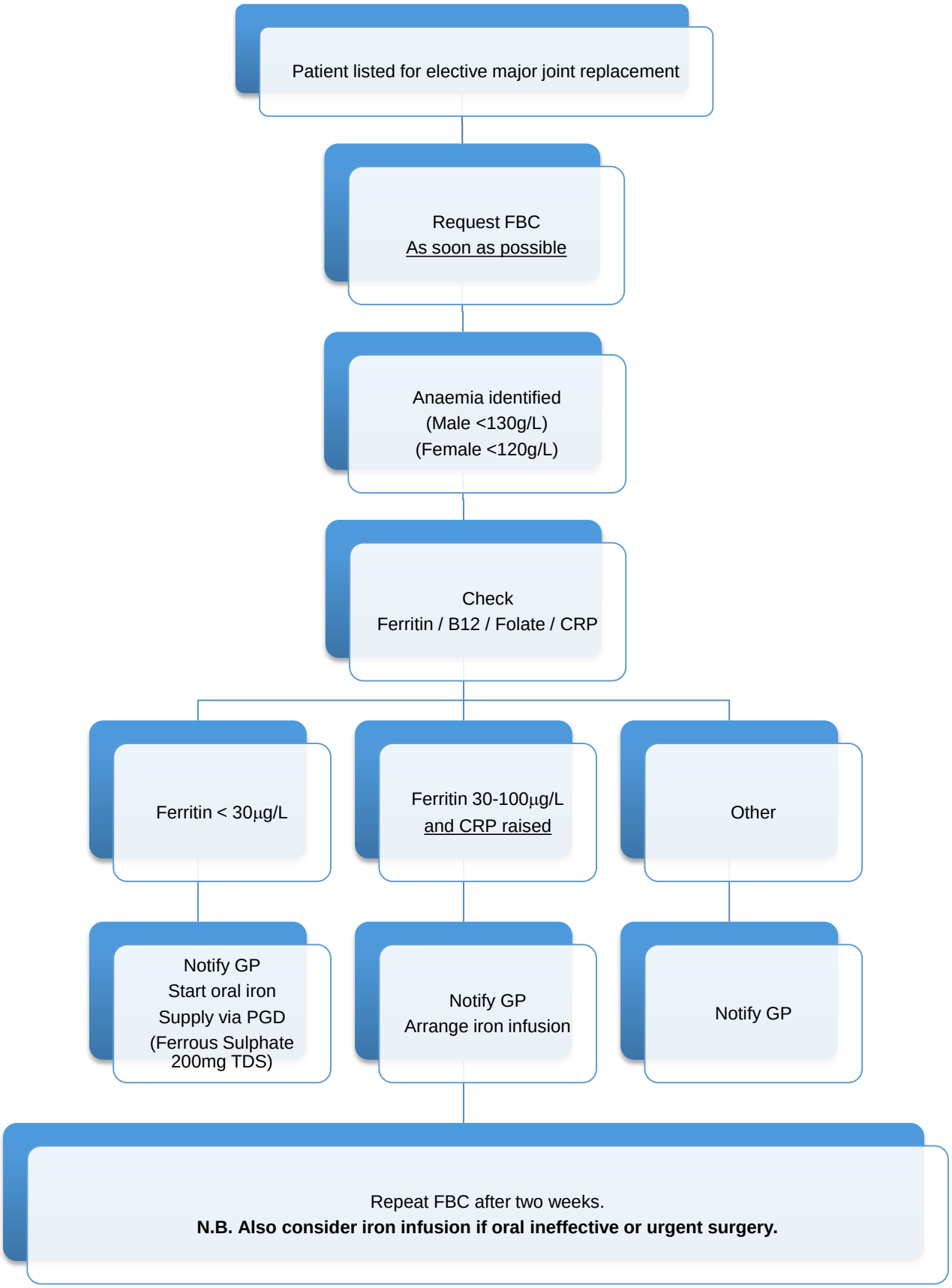
5.2. The flowchart in Appendix 6.1 will be utilised by the Pre-op assessment nurses to initiate appropriate patient management and the standard letters in Appendix 6.2 will be used by them to ensure prompt and accurate communication with primary care.

5.3. The process will be monitored internally within the Pre-op Assessment team and, more broadly, as a component part of the Patient Blood Management initiative via the Hospital Transfusion Committee

6. Appendices

6.1. Flowchart for the management of pre-operative anaemia (see next page)

Management of Anaemia Prior to Elective Major Joint Replacement



6.2. Standard letters for communication with primary care (see next page)

6.2.1. GP letter for use with oral iron

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Dear Doctor,

Re.

Patient Label Here

The above patient has been identified as being anaemic during their preparation for surgery.

- They have been started on oral iron due to evidence of iron deficiency (Ferritin <30μg/L)

Please consider any further investigations as necessary.

We have attached a copy of our pre-operative anaemia pathway by way of further explanation.

Kind regards,

Pre-Op Assessment Nurse
North Devon District Hospital

Chief Executive: Dr Alison Diamond

Chair: Roger French

6.2.2. GP letter for use with iron infusion

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www.northdevonhealth.nhs.uk

Dear Doctor,

Re.

Patient Label Here

The above patient has been identified as being anaemic during their preparation for surgery.

- An iron infusion has been arranged on the grounds of either:
  - Ferritin <30µg/L
  - Ferritin 30-100µg/L with elevated CRP

Please review their B12/folate results (if required commence treatment immediately) and consider any further investigations as necessary.

We have attached a copy of our pre-operative anaemia pathway by way of further explanation.

Kind regards,

Pre-Op Assessment Nurse
North Devon District Hospital

Chief Executive: Dr Alison Diamond

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6.2.3. GP letter for use in other anaemias (i.e. no iron therapy)

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Dear Doctor,

Re.

Patient Label Here

The above patient has been identified as being anaemic during their preparation for surgery.

- Results suggest that they will not benefit from iron therapy pre-operatively on the basis of either:
  - Ferritin >100μg/L
  - Ferritin >30μg/L with normal CRP

Please review their B12/folate results (if required commence treatment immediately) and consider any further investigations as necessary.

We have attached a copy of our pre-operative anaemia pathway by way of further explanation.

Kind regards,

Pre-Op Assessment Nurse
North Devon District Hospital

Chief Executive: Dr Alison Diamond

Chair: Roger French

7. References

- British Committee for Standards in Haematology Guidelines on the Identification and Management of Pre-Operative Anaemia. *British Journal of Haematology*, 2015, **171**, 322-331
- International consensus statement on the peri-operative management of anaemia and iron deficiency. *Anaesthesia*, 2017, **72**, 233-247

8. Associated Documentation

- Iron Replacement Therapy Assessment of Clinical Competency, Northern Devon Healthcare Trust
- Patient Group Direction for the Supply of Ferrous Sulphate 200mg tablets by Registered Practitioners employed by Northern Devon Healthcare NHS Trust